

# PhyGNN-CPDD: Physics-Informed GNN Framework for Protein Pocket Discovery

Main Prediction Pipeline →

Hamiltonian Physics Constraints ( $\lambda=0.0001$ )

**Protein Structure**  
PDB File  
(1a0q, 1a2c, ...)

**Graph Construction**  
Nodes: Residues  
Edges: Spatial/Sequence

**Hamiltonian GNN**  
Physics-Informed Graph Neural Network

GNN Layers    Attention Mechanism    Physics Layer

**Pocket Prediction**  
Binary Classification  
Confidence Scores

**Explainability**  
GNNExplainer  
SHAP Analysis

**Electrostatics**

**van der Waals**

**Bond Lengths**

**Angles**

**Hydrogen Bonds**

$\sum (q_i q_j) / (4\pi\epsilon_0 r_{ij})$

**Structural Features (20 dimensions)**

- Residue Type
- Secondary Structure
- Solvent Accessibility

$4\epsilon[(\sigma/r)^{12} - (\sigma/r)^6]$

$\frac{1}{2}k_a(r - r_0)^2$

**Physics Features (8 dimensions)**

- Partial Charges
- VDW Radii
- Hydrogen Bonds
- Hydrophobicity

$\frac{1}{2}k_\theta(\theta - \theta_0)^2$

$E = \epsilon[5(\sigma/r)^{12} - 6(\sigma/r)^{10}]$

**Geometric Features (7 dimensions)**

- 3D Coordinates
- Distances
- Angles
- Dihedrals

35-Dimensional Feature Representation

Physics constraints integrated via Hamiltonian loss:  $L_{total} = L_{pred} + \lambda \cdot L_{physics}$